

DA5400W

Tutorial - 1

Question

The following probability distribution function can describe the growth rate of finance:

$$f(x) = \frac{1}{\alpha^2} x e^{-x/\alpha}$$

with

$$\alpha \in (0, \infty), \quad x \in [0, \infty)$$

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- Maximum Likelihood Estimation (MLE)
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Given data:

$$x_1, x_2, \dots, x_n$$

Step 1: Likelihood Function (MLE)

For independent observations:

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Step 2: Log-Likelihood

Taking logarithm:

$$\ell(\alpha) = \ln L(\alpha)$$

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$$\ell(\alpha) = \ln L(\alpha)$$

$$\ell(\alpha) = -2n \ln \alpha + \sum_{i=1}^n \ln x_i - \frac{1}{\alpha} \sum_{i=1}^n x_i$$

Step 3: Differentiate

Differentiate w.r.t. α :

$$\frac{d\ell}{d\alpha} = -\frac{2n}{\alpha} + \frac{1}{\alpha^2} \sum_{i=1}^n x_i$$

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$$\frac{d\ell}{d\alpha} = -\frac{2n}{\alpha} + \frac{1}{\alpha^2} \sum_{i=1}^n x_i$$

Set derivative equal to zero:

$$-\frac{2n}{\alpha} + \frac{1}{\alpha^2} \sum_{i=1}^n x_i = 0$$

Step 4: Solve for α (MLE)

Multiply by α^2 :

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$$\hat{\alpha}_{\text{MLE}} = \frac{1}{2n} \sum_{i=1}^n x_i$$

Method of Moments: Step 1

Population mean is:

$$\mathbb{E}[X] = \int_0^{\infty} xf(x) dx$$

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$$\mathbb{E}[X] = \frac{1}{\alpha^2} \int_0^{\infty} x^2 e^{-x/\alpha} dx$$

Method of Moments: Step 2

Using Gamma integral:

$$\int_0^{\infty} x^2 e^{-x/\alpha} dx = 2! \alpha^3$$

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$$\int_0^{\infty} x^2 e^{-x/\alpha} dx = 2! \alpha^3$$

$$\mathbb{E}[X] = \frac{1}{\alpha^2} (2\alpha^3) = 2\alpha$$

Method of Moments: Step 3

Sample mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

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Conclusion: Both MLE and Method of Moments give the same estimator.

Question

A random sample

$$X_1, X_2, \dots, X_9$$

is drawn from a population with mean μ and variance σ^2 .

Two estimators are proposed:

$$\theta_{\text{Mr}} = \frac{X_1 + X_3 + 4X_5}{6}$$

$$\theta_{\text{Mrs}} = \frac{X_2 - X_6 + 2X_7 + 3X_4}{5}$$

Question: Which estimator is the best and why?

Approach

To decide the best estimator, we will check:

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If both estimators are unbiased, the one with **smaller variance** is preferred.

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Key idea: Among unbiased estimators, the one with minimum variance is the **best**.

Step 1: Unbiasedness of θ_{Mr}

$$\mathbb{E}[\theta_{Mr}] = \frac{1}{6} (\mathbb{E}[X_1] + \mathbb{E}[X_3] + 4\mathbb{E}[X_5])$$

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$$\begin{aligned}\mathbb{E}[\theta_{Mr}] &= \frac{1}{6} (\mathbb{E}[X_1] + \mathbb{E}[X_3] + 4\mathbb{E}[X_5]) \\ &= \frac{1}{6} (\mu + \mu + 4\mu)\end{aligned}$$

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$$\mathbb{E}[\theta_{Mr}] = \mu$$

θ_{Mr} is unbiased.

Step 2: Unbiasedness of θ_{Mrs}

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$$\text{Var}(\theta_{Mr}) = \frac{\sigma^2}{2}$$

Step 4: Variance of θ_{Mrs}

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$$\text{Var}(\theta_{Mrs}) = \frac{1}{25}(\sigma^2 + \sigma^2 + 4\sigma^2 + 9\sigma^2)$$

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Comparison and Conclusion

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$$\text{Var}(\theta_{Mr}) < \text{Var}(\theta_{Mrs})$$

θ_{Mr} is the best estimator

Because both estimators are unbiased and θ_{Mr} has smaller variance.

Problem:

Consider a random variable C with mean μ and standard deviation σ . Two experiments are performed to collect two random samples with sample sizes n_1 and n_2 .

Let the sample means be $\bar{C}_1$ and $\bar{C}_2$ respectively.

Show that the estimator

$$\bar{C} = a\bar{C}_1 + (1 - a)\bar{C}_2$$

is an unbiased estimator for the population mean μ .

To check whether an estimator is **unbiased**, we compute its expectation.

If

$$\mathbb{E}[\bar{C}] = \mu$$

then the estimator is unbiased.

Expectation of Sample Means

For any random sample mean:

$$\mathbb{E}[\bar{C}_1] = \mu$$

$$\mathbb{E}[\bar{C}_2] = \mu$$

This holds regardless of the sample sizes n_1 and n_2 .

Expectation of the Estimator

Take expectation of the estimator:

$$\mathbb{E}[\bar{C}] = \mathbb{E} [a\bar{C}_1 + (1 - a)\bar{C}_2]$$

Using Linearity of Expectation

Using linearity of expectation:

$$\mathbb{E}[\bar{C}] = a \mathbb{E}[\bar{C}_1] + (1 - a) \mathbb{E}[\bar{C}_2]$$

Substitute the expectations of sample means:

$$\mathbb{E}[\bar{C}] = a\mu + (1 - a)\mu$$

$$\mathbb{E}[\bar{C}] = \mu(a + 1 - a)$$

$$\mathbb{E}[\bar{C}] = \mu$$

$$\mathbb{E}[\bar{C}] = \mu$$

Therefore,

$$\bar{C} = a\bar{C}_1 + (1 - a)\bar{C}_2$$

is an **unbiased estimator** of the population mean μ .

Determine the stationary points and classify their nature for the function:

$$f(x, y) = x^4 + y^4 - 36xy$$

Approach

To find stationary points:

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$$D = f_{xx}f_{yy} - (f_{xy})^2$$

- $D > 0, f_{xx} > 0 \Rightarrow$ local minimum

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- $D > 0, f_{xx} > 0 \Rightarrow$ local minimum
- $D > 0, f_{xx} < 0 \Rightarrow$ local maximum
- $D < 0 \Rightarrow$ saddle point

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$$x(x^8 - 6561) = 0$$

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Corresponding y values:

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Corresponding y values:

$$(0, 0), \quad (3, 3), \quad (-3, -3)$$

Step 3: Hessian and Classification

Second derivatives:

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$$f_{xx} = 12x^2, \quad f_{yy} = 12y^2, \quad f_{xy} = f_{yx} = -36$$

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Check each point:

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Check each point:

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$$D = 0 * 0 - (-36)^2 = -1296 < 0 \Rightarrow \text{Saddle Point}$$

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2. At $(3, 3)$:

$$D = 108 * 108 - 36^2 = 10368 > 0, \quad f_{xx} = 108 > 0 \Rightarrow \text{Local Minimum}$$

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